Muhammad Haris Ali

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Professional Profile

AI Engineer with expertise in Generative AI, Large Language Models, Computer Vision, and Machine Learning, experienced in designing, developing, and deploying intelligent AI solutions for government and enterprise applications. Skilled in building production-ready AI systems, fine-tuning large language models, developing computer vision pipelines, and optimizing deep learning models for high-performance GPU inference. Proficient in Python, PyTorch, TensorFlow, Docker, Kubernetes, CUDA, and TensorRT, with hands-on experience across the complete AI lifecycle from research and model development to deployment, optimization, and MLOps. Passionate about delivering secure, scalable, and impactful AI solutions that address complex real-world challenges.

Education

- **Bahria University** — Islamabad, Pakistan
  Bachelor of Science in Computer Science — *2021 – 2025*

Professional Experience

- **Government of Pakistan** — Islamabad, Pakistan
  AI Engineer (BPS-17) — *Sep 2025 – Present*
  - Developing and deploying AI-driven solutions for national-scale digital transformation initiatives within a secure government environment.
  - Designing Computer Vision, Machine Learning, Natural Language Processing (NLP), and Large Language Model (LLM) based applications for mission-critical use cases.
  - Working on confidential government-level AI projects involving intelligent automation, document intelligence, decision support, and secure AI systems.
  - Building scalable AI services, REST APIs, and microservices for production deployment while ensuring security, reliability, and maintainability.
  - Optimizing deep learning models using TensorRT, ONNX Runtime, CUDA, and GPU acceleration techniques to achieve high-performance inference.
  - Setting up and maintaining local, on-premises GPU infrastructure for AI model deployment, including NVIDIA driver and CUDA Toolkit installation, system-level environment configuration, and GPU health monitoring.
  - Containerizing AI workloads with Docker and orchestrating multi-service deployments using Kubernetes for scalable, on-premises inference pipelines.
  - Configuring NVIDIA Multi-Process Service (MPS) and GPU partitioning strategies to enable efficient CUDA workload sharing and concurrent model serving on shared hardware.
  - Collaborating with multidisciplinary engineering teams throughout the complete AI lifecycle including data engineering, model development, deployment, monitoring, and continuous optimization.
- **Techostudios** — Remote
  Generative AI Engineer | Financial AI Product — *Dec 2024 – Sep 2025*
  - Architected an end-to-end AI platform for financial forecasting and transaction analysis using deep learning and Generative AI.
  - Developed LSTM-based forecasting models for equity trend prediction and anomaly detection across large-scale financial datasets.
  - Engineered time-series feature pipelines, including rolling statistics, technical indicators, and sequence windowing, to improve model signal quality and reduce noise in volatile market data.
  - Integrated Large Language Models to generate automated financial insights and analytical reports.
  - Designed prompt templates and retrieval workflows to ground LLM-generated summaries in verified transaction and market data, improving factual consistency of automated reports.
  - Managed the complete ML lifecycle including data preprocessing, feature engineering, model training, evaluation, deployment, and API integration.
  - Containerized model training and inference services with Docker to standardize environments across development and production.
- **Freelance** — Remote
  Generative AI Engineer — *May 2023 – Present*
  - Developed AI-powered interview assistants with speech recognition, LLM reasoning, and automated performance evaluation.
  - Built synthetic data generation pipelines using Large Language Models for healthcare and enterprise NLP applications.
  - Applied quality-filtering and diversity-sampling heuristics to synthetic datasets to reduce bias and improve downstream model generalization.
  - Designed multimodal semantic search systems using CLIP embeddings, vector databases, and Retrieval-Augmented Generation (RAG).
  - Developed AI-driven resume screening systems using transformer-based architectures for intelligent candidate ranking.

- Built real-time Computer Vision solutions using YOLO for edge deployment on NVIDIA Jetson platforms.
- Optimized edge inference pipelines with model quantization and TensorRT conversion to reduce latency and memory footprint on resource-constrained devices.
- Designed enterprise Retrieval-Augmented Generation (RAG) systems integrating vector databases, document processing, and modern LLM orchestration frameworks.
- Set up local development and deployment environments using Docker and CUDA-enabled containers to ensure reproducible GPU-accelerated workflows across client projects.

- **RISETech, NUST College of E&ME** — Rawalpindi, Pakistan
  Machine Learning Intern — *Jul 2024 – Aug 2024*
  - Fine-tuned Large Language Models including LLaMA and Gemma for healthcare and educational applications.
  - Applied parameter-efficient fine-tuning techniques to adapt pretrained LLMs to clinical dialogue tasks with limited computational resources.
  - Developed an AI-powered Medical OSCE simulation system supporting history taking, diagnosis, and automated performance assessment.
  - Implemented custom datasets, prompt engineering strategies, and evaluation pipelines to improve model performance for clinical reasoning tasks.
  - Collaborated with faculty supervisors to validate model outputs against clinical accuracy benchmarks and iterated on training data quality.

Projects

- **VetAI Chatbot (Final Year Project)**
  Multi-Module Generative AI | Flutter | LLM Fine-Tuning | Voice AI
  - Built a multi-module AI chatbot for virtual veterinary services with a Flutter frontend and three integrated AI engines.
  - Fine-tuned LLaMA for accurate disease diagnosis and treatment suggestions in veterinary contexts.
  - Integrated Whisper for Urdu voice communication, improving rural accessibility.
  - Used Gemini for conversational flow management and structured medical report generation.
- **ANPR Gate Entry System (Jetson Nano Edge AI Deployment)**
  *Computer Vision | YOLOv8 | Edge Processing | **2nd Position – Project Gala 2023***
  - Deployed an Automatic Number Plate Recognition system on Jetson Nano for real-time edge inference.
  - Trained custom YOLOv8 model on self-curated Pakistani number plate dataset with OCR post-processing.
  - Achieved low-latency on-device inference with no cloud dependency.
  - Awarded 2nd position at Project Gala 2023 for innovation in embedded AI.
- **HACRF Spectrum Analyzer for Drone Detection**
  Edge AI | Signal Processing | ARM/x64/AMD | Confidential Client
  - Developed a spectrum analysis solution for drone detection, processing time-varying RF frequency data.
  - Built hardware-specific implementations targeting ARM, x64, and AMD architectures for edge processing.
  - Implemented algorithms to detect characteristic frequency signatures of drones with low false-positive rates.
  - Designed for autonomous operation in contested environments with minimal latency.
- **Synthetic Data Generation Pipeline for Healthcare NLP**
  LLM | Data Augmentation | Healthcare AI
  - Created a pipeline using LLMs to generate 50K+ synthetic labeled text samples for healthcare NLP tasks, reducing manual annotation costs by 80%.
  - Implemented quality filtering and diversity sampling to ensure dataset quality and coverage.
  - Used generated data to improve downstream classifier accuracy by 15% over baseline.
- **Cancer Cell Segmentation & Classification**
  Computer Vision | U-Net | Medical Imaging | Deep Learning
  - Developed a deep learning pipeline for automated cancer cell segmentation using U-Net architecture.
  - Benchmarked against ResNet and EfficientNet classifiers for cell-type identification.
  - Applied data augmentation, transfer learning, and class imbalance handling techniques.
- **SAM 3D: Sketch-to-Animation for Toddlers**
  Computer Vision | Segment Anything Model | 3D Rendering | EdTech
  - Built an interactive learning tool converting children's sketches into animated 3D characters.
  - Used Meta's Segment Anything Model for precise sketch segmentation.
  - Implemented sketch-to-3D pipeline with model rigging and procedural animation.
- **RAG System for Enterprise Knowledge Bases**
  LLM | Retrieval-Augmented Generation | Vector Search
  - Developed a RAG pipeline for enterprise document retrieval and Q&A, integrating ingestion, chunking, vector embeddings, and LLM orchestration.
  - Deployed using LangChain, OpenAI embeddings, and Pinecone vector database with sub-second query latency.

- Tuned chunking strategy and retrieval top-k parameters to balance answer relevance against response latency for enterprise-scale document sets.

- **Document Summarization Tool**
  NLP | TF-IDF | PyMuPDF | Flask Web App
  - Developed web application for PDF text extraction and summarization using TF-IDF scoring with NLTK.
  - Built a Flask-based interface allowing users to upload PDFs and receive extractive summaries with adjustable compression ratio.
- **Real-Time Inventory Detection with YOLOv8 on Jetson**
  Computer Vision | YOLOv8 | Edge AI | Retail Automation
  - Built an edge-optimized inventory management system using YOLOv8 deployed on NVIDIA Jetson Nano for real-time object detection and stock monitoring.
  - Optimized model inference with TensorRT conversion and batch processing to maintain real-time frame rates on constrained edge hardware.

Skills

- **AI & Deep Learning:** LLM Fine-Tuning, Computer Vision, LSTM, U-Net, YOLOv8, SAM, Generative AI, NLP, Time-Series Forecasting, Signal Processing, Transformer Architectures, Attention Mechanisms, Retrieval-Augmented Generation (RAG), Stable Diffusion, CLIP, LoRA, Embeddings
- **ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, spaCy, NLTK, OpenCV, FastAI, XGBoost,
- **Model Optimization:** TensorRT, ONNX, OpenVINO, CUDA, CUDA Slicing, MPS, Model Pruning, Quantization, Knowledge Distillation, GPU Acceleration
- **DevOps & MLOps:** Docker, Kubernetes, NVIDIA Container Toolkit, AWS (EC2, S3), GCP, MinIO, FastAPI, Flask, REST APIs, Microservices
- **Local AI Deployment & GPU Systems:** NVIDIA CUDA Toolkit Setup, NVIDIA Drivers, CUDA MPS (Multi-Process Service), GPU Partitioning, nvidia-smi, On-Premises GPU Infrastructure Configuration
- **Programming Language:** Python, C++, PHP, Laravel, Bash
- **Databases:** SQL, PostgreSQL, MySQL, MongoDB, Redis,FAISS
- **Platforms:** Linux, Git, GitHub Actions, MLflow, Kubeflow, Jupyter, NVIDIA Jetson Nano, Raspberry Pi

Certifications

- **Google Data Analytics Professional Certificate** – Google
- **AWS Academy Cloud Architecting** – Amazon Web Services
- **Introduction to Data Science** – IBM
- **AI for Everyone** – Google
- **Agentic AI – Hands-On Training** – Agentic workflows, autonomous AI systems, and tool-use patterns
- **Deep Learning Specialization** – Coursera (In Progress)

Core Competencies

- **Technical Leadership:** Owning AI systems end-to-end from research and prototyping through production deployment and monitoring.
- **Cross-Functional Collaboration:** Working with data engineers, backend teams, and stakeholders to translate requirements into deployable AI solutions.
- **Problem Solving:** Diagnosing performance bottlenecks in training and inference pipelines and applying targeted optimization strategies.
- **Communication:** Translating technical AI concepts into clear insights and recommendations for non-technical stakeholders.

Achievements

- **2nd Position – AI, Cyber & Drone Swarm Gala** – Awarded for a proof-of-concept facial image enhancement system that processes CCTV footage of drivers and co-passengers inside vehicles to improve image quality for downstream facial recognition.
- **3nd Position – Project Gala 2023**, Bahria University – Awarded for the ANPR Gate Entry System, an embedded edge-AI solution deployed on NVIDIA Jetson Nano.

Languages

- **English:** Professional Working Proficiency
- **Urdu:** Native